

8. (2.75 points)

Given the following information:

| Observed Age-to-Age Loss Development Factors<br>as of (years) |       |       |       |       |
|---------------------------------------------------------------|-------|-------|-------|-------|
| AY                                                            | 1-2   | 2-3   | 3-4   | 4-5   |
| 1                                                             | 1.700 | 1.350 | 1.100 | 1.050 |
| 2                                                             | 2.500 | 1.550 | 1.080 |       |
| 3                                                             | 2.000 | 1.400 |       |       |
| 4                                                             | 1.800 |       |       |       |

- There is no development after the 5<sup>th</sup> period.
- Probability of a standard normal variate lying in the interval  $(-0.67, 0.67)$  is 50%.

a. (2 points)

Evaluate whether subsequent columns of development factors exhibit correlation using Spearman's rank at a 50% confidence threshold.

b. (0.5 point)

Briefly describe two reasons why it is more appropriate to test the triangle as a whole rather than correlation between pairs of columns.

c. (0.25 point)

Identify which assumption of the chain ladder method is being evaluated in part a. above.

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**SPRING 2018 EXAM 7, QUESTION 8**

**TOTAL POINT VALUE: 2.75**

**LEARNING OBJECTIVE(S): A2**

**SAMPLE ANSWERS**

**Part a: 2 points**

Sample 1

$$S_k = \sum (rank_k - rank_{k-1})^2; T_k = 1 - 6 \times \frac{S_k}{n \times (n^2 - 1)}; W_k = \#AY - k - 1$$

k=2, rank

|            |            |
|------------|------------|
| <u>1-2</u> | <u>2-3</u> |
| 1          | 1          |
| 3          | 3          |
| 2          | 2          |

$$S_2 = 0; T_2 = 1; W_2 = 5 - 2 - 1 = 2$$

k=3, rank

|            |            |
|------------|------------|
| <u>2-3</u> | <u>3-4</u> |
| 1          | 2          |
| 2          | 1          |

$$S_3 = 1^2 + 1^2; T_2 = 1 - \frac{2}{2 \times \frac{3}{6}} = -1; W_3 = 1$$

$$T = \frac{\sum T_k \times W_k}{\sum W_k} = \frac{1 \times 2 + (-1) \times 1}{2 + 1} = 0.333$$

$$Var(T) = \frac{1}{\frac{(\#AY - 2) \times (\#AY - 3)}{2}} = \frac{1}{\frac{(5 - 2) \times (5 - 3)}{2}} = 0.333$$

$$C.I. at 50\% = (\sqrt{0.333} \times (-0.67), \sqrt{0.333} \times 0.67) = (-0.387, 0.387)$$

Since T = 0.333 falls into C.I. at 50%, do not reject null hypothesis. Hence, no correlation is observed.

Sample 2

| AY | r <sub>1</sub> | r <sub>2</sub> | S <sub>2</sub> | r <sub>3</sub> | S <sub>3</sub> |
|----|----------------|----------------|----------------|----------------|----------------|
| 1  | 1              | 1              | 1              | 1              | 2              |
| 2  | 4              | 3              | 3              | 2              | 1              |
| 3  | 3              | 2              | 2              |                |                |
| 4  | 2              |                |                |                |                |

$$I = 5$$

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$$T = 1 - 6 \times \frac{\sum(r - s)^2}{n^3 - n}$$

$$T_2 = 1 \text{ (because perfect correlation, positive)}$$

$$T_3 = -1 \text{ (because perfect correlation, negative)}$$

$$T_{global} = \frac{2 \times 1 + 1 \times (-1)}{3} = 0.\bar{3}$$

$$E(T) = 0, Var(T) = \frac{1}{\frac{(I-3) \times (I-2)}{2}} = \frac{1}{\frac{(5-3) \times (5-2)}{2}} = 1/3$$

$$interval = (\sqrt{0.333} \times (-0.67), \sqrt{0.333} \times 0.67) = (-0.3866, 0.3866)$$

T = 1/3 is between (-0.3866, 0.3866), so columns of development do not exhibit correlation.

Sample 3

| AY | r <sub>1</sub> | r <sub>2</sub> | s <sub>2</sub> | r <sub>3</sub> | s <sub>3</sub> |
|----|----------------|----------------|----------------|----------------|----------------|
| 1  | 1              | 1              | 1              | 1              | 2              |
| 2  | 4              | 3              | 3              | 2              | 1              |
| 3  | 3              | 2              | 2              |                |                |
| 4  | 2              |                |                |                |                |

$$I = 5$$

$$T_2 = 1 - 6 \times \frac{\sum(r_2 - s_2)^2}{3^3 - 3} = 1$$

$$T_3 = 1 - 6 \times \frac{1^2 + 1^2}{2^2 - 2} = -1$$

$$T = \frac{2 \times 1 + 1 \times (-1)}{(5 - 2 - 1) + (5 - 3 - 1)} = \frac{1}{1 + 2} = 0.333$$

$$E(T) = 0, Var(T) = \frac{1}{3 \times \frac{2}{2}} = 0.333$$

$$interval = (\sqrt{0.333} \times (-0.67), \sqrt{0.333} \times 0.67) = (-0.3866, 0.3866)$$

$$C.I. = \pm 0.67 \times \sqrt{\frac{1}{3}} = \pm 0.3868$$

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0.33 Falls within the CI, so we conclude that there is not significant correlation at 50% level.

Sample 4

$$I = 5 = \#AY; S_k = \sum (rank - rank_{prior})^2; T_k = 1 - \frac{S_k}{n \times \frac{(n^2 - 1)}{6}}; Wt = I - k - 1$$

|       |     |     |
|-------|-----|-----|
|       | 2-2 | 3-4 |
| k     | 2   | 3   |
| $S_k$ | 0   | 2   |
| $T_k$ | 1   | -1  |
| n     | 3   | 2   |
| wt    | 2   | 1   |

$$T = \frac{\sum wt \times T_k}{\sum wt} = \frac{2 \times 1 + 1 \times (-1)}{2 + 1} = 0.333$$

$$E(T) = 0, Var(T) = \frac{1}{\frac{(I-2) \times (I-3)}{2}} = 0.333$$

$$CI = E(T) \pm \sqrt{V(T)} \times 0.67 = (-0.387, 0.387)$$

T = 0.333 is within interval, no subsequent correlation exists.

Sample 5

| <u>AY</u> | <u>1-2 rank</u> | <u>2-3 rank</u> |
|-----------|-----------------|-----------------|
| 1         | 1               | 1               |
| 2         | 3               | 3               |
| 3         | 2               | 2               |

*Perfectly correlated*  $\Rightarrow \rho_1 = 1$

| <u>AY</u> | <u>1-2 rank</u> | <u>2-3 rank</u> |
|-----------|-----------------|-----------------|
| 1         | 1               | 2               |
| 2         | 2               | 1               |

*Perfect negative correlation*  $\Rightarrow \rho_2 = -1$

Combine, weighted by # obs - 1

$$\rho_{overall} = \frac{2 \times 1 + 1 \times (-1)}{3} = 0.333$$

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$$\text{Var}(\rho_{\text{overall}}) = \left(\frac{2}{3}\right)^2 \times \left(\frac{1}{2}\right) + \left(\frac{1}{3}\right)^2 \times 1 = 0.333, E(\rho_{\text{overall}}) = 0$$

$$Z = \frac{\rho_{\text{overall}} - E(\rho_{\text{overall}})}{\sqrt{\text{Var}(\rho_{\text{overall}})}} = \frac{0.333 - 0}{\sqrt{0.333}} = 0.5774$$

Since Z is in (-0.67, 0.67), we do not reject the hypothesis that columns are uncorrelated.

### Part b: 0.5 point

#### Sample Response for avoiding an accumulation of error probabilities

- Avoid an accumulation of error probabilities.
- It avoids the accumulation of error probabilities. In other words, we have a smaller chance of making a type 1 error.
- It's not unusual to see correlation in a few pairs of columns, could be due to randomness but doesn't necessarily mean the whole triangle has correlation.
- Naturally there is a chance that a few columns happen to appear correlated, we cannot say if it is really a correlation or because of randomness.
- Reduce the probability of making errors.
- Due to randomness, we may have some pairs of LDFs exhibiting correlation although overall they are not correlated.
- For each pair, even at 90% confidence level there is still a 10% chance that the pair randomly shows correlation.
- Given a large triangle, there will be correlation between columns by pure chance.

#### Sample Response for the importance of knowing whether correlations globally prevail

- More important to know whether correlations globally prevail than to find a small part of the triangle with correlations.
- The whole triangle is more important than a pair of columns.
- We focus more on the total unpaid or unreported losses confidence interval, so we should focus more on the entire triangle.
- Because we use the triangle as a whole for valuation.
- Correlation in the whole triangle is more meaningful when doing analysis.
- We care about whether there is development factor correlation as a whole rather than a localized region.
- It's better to account for correlation in the entire dataset than in a small section of the dataset.
- Testing the triangle as a whole gives an overview of the correlations exhibited in the data, which makes more intuitive sense and provides an overall conclusion.

### Part c: 0.25 point

#### Sample Response

- The expected cumulative losses in the next development period is proportional to losses up-to-date.
- Incremental losses in a period only depend on cumulative losses in a prior period.

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- $E[C_{i,k+1}|C_{i,1}, \dots, C_{i,k}] = C_{i,k} \times f_k$
- $E[C_{i,k+1}|data] \propto C_{i,k}$
- Accident years are independent.
- $C(i,k)$  and  $C(j,l)$  are independent for all  $i \neq j$  and  $k \neq l$ .
- Development years and accident years are independent.

### EXAMINER'S REPORT

Candidates were expected to understand the linearity assumption of the chain ladder method, how to perform a test of the assumption using Spearman correlation, and why it is more appropriate to test the entire triangle globally rather than all column pairs individually.

### Part a

Candidates were expected to perform a correlation test using Spearman correlation coefficients and draw a conclusion as to whether subsequent columns of development factors exhibited correlation. This included:

- Calculating the Spearman correlation coefficients between the first two pairs of development factors.
- Taking a weighted average of these correlation coefficients to serve as a test statistic.
- Calculating the confidence interval for the test statistic under the null hypothesis that columns are uncorrelated, which requires calculating the variance of the test statistic and knowing that the expected value of the test statistic is zero.
- Comparing the test statistic to the confidence interval to draw a conclusion on the null hypothesis, and stating the hypothesis being tested.

Common mistakes included:

- Evaluating the correlation between each column pair individually instead of deriving a single test statistic for the triangle, as the syllabus material stresses that simply testing pairs individually is suboptimal.
- Performing a test that relies on Pearson correlation coefficients even though the problem explicitly states the use of Spearman correlation coefficients.
- Using the wrong value for  $l$ .
- Skipping the variance calculation and using the standard normal variate directly without normalizing the test statistic.
- Using an incorrect equation for the confidence interval of  $T$  (e.g. multiplying 0.67 by the variance of  $T$  instead of the standard deviation of  $T$ , or using 0.5 from the confidence interval instead of the standard normal variate of 0.67).
- Errors in ranking the development factors.
- Misstating the conclusion, including stating that the null hypothesis was rejected instead of not rejected or that the null hypothesis was that factors are correlated.
- Performing the wrong test, including a calendar year effects test or a correlation test based on Pearson correlation coefficients.
- Simply stating "do not reject null" as the conclusion without stating what the null hypothesis is or whether this means that correlation is/isn't present.

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### Part b

Candidates were expected to describe two reasons why it is more appropriate to test the triangle as a whole, rather than individually testing all pairs in the triangle. Candidates were expected to provide reasons in line with two general themes:

- Testing all pairs individually creates an accumulation of type 1 error probabilities.
- It is more important to know whether correlations globally prevail than to find a small part of the triangle with correlations.

Common mistakes included:

- Providing a second reason that was a rewording of the first.
- Arguing against testing a *single column pair* versus the whole triangle, rather than against testing *pairs of columns* versus the whole triangle.
- Stating that correlation may exist between non-subsequent columns, since part (b) of the question asks about “correlation between pairs of columns” and does not restrict this to subsequent pairs.
- Providing issues that would not be addressed by a development factor correlation test (individually or globally) and would require a different test, such as calendar year effects.
- Providing causes of correlation of calendar year effects such as “larger developments tend to be followed by smaller developments” or “a change in reserving practices or settlement rates” rather than a reason for performing a test on the triangle as a whole instead of testing pairs individually.

### Part c

Candidates were expected to know that the chain ladder method assumption being evaluated is the linearity assumption (i.e. expected claims in a period only depend on the previous level of cumulative claims and a factor based on age).

However, it was noted that there is a conflict in the syllabus material on this subject, with some of the material stating that a development factor correlation test evaluates the assumption of accident year independence. As a result, candidates received credit for either response.

Common mistakes included:

- Providing the variance assumption of the chain ladder method.
- Simply stating “the linearity assumption” without elaboration to demonstrate understanding of what the assumption entails.
- Stating that the test in part a. evaluates the assumption that development factors are uncorrelated, as this is an implication of the linearity assumption of the chain ladder method rather than its own assumption.
- Writing an incomplete mathematical expression without adding additional verbal description to complete the assumption (e.g. simply writing “ $E[C_{k+1}] = C_k \times f_k$ ”, as this omits that the expected value is conditional on the previous level of cumulative claims).